

Three Dimensional Protein Structures Alignments by Minimum Bipartite Matching

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Abstract

Protein structure provides the opportunity to recognize homology that is undetectable by sequence comparison, and it represents a powerful means of discovering functions, yielding direct insight into the molecular mechanisms. Currently, there are several techniques available in attempting to find the optimal alignment of shared structural motifs between two proteins.

In this paper, we propose algorithms and develop tools for pairwise alignment of protein structures. Methods of locating suitable isometric transformations of one structure and aligning it to the other structure are addressed. Our methods allow sequence gaps of any length, reversal of chain direction, and free topological connectivity of atom sequences. We show the effectiveness of the proposed refinement methods by a set of experiments, which improve several previous results.

Keywords: structural proteomics, algorithms, structure alignments and comparisons, *rmsd*, minimum bipartite matching.

1 Introduction

One of the primary goals of structural alignment programs is to quantitatively measure the level of structural similarity between all pairs of known protein structures. This data can provide several meaningful insights into the nature of protein structures and their functional mechanisms.

The three dimensional structure of proteins is highly conserved during evolution [4]. Protein are constructed by one or more polypeptide chains

that fold into complicated 3D structures. Detection of proteins with a similar fold can suggest a common ancestor, and often a similar function [7]. Comparison of 3D structures makes it possible to establish distant relationships, even between protein families distinct in terms of sequence comparison alone. This is why structural alignment of proteins increases our understanding of more distant evolutionary relationships [3, 13]. The link between structural classification and sequence families enables us to study functions of various folds, or whole proteins [15].

The smallest *root mean squared deviation* (*rmsd*) is a least-squares fitting method for two sequences of points [12]. The idea is to align atom vectors of the two given (molecular) structures, and use the common least averaged squared errors as a measurement of differences between these two (paired) sequences. Formally, let $P = \langle p_1, \dots, p_n \rangle$ and $Q = \langle q_1, \dots, q_n \rangle$ be two sequences of points. We assume that P is translated so that its centroid ($\frac{1}{n} \sum_{k=1}^n p_k$) is at the origin. We also assume that Q is translated in the same way. For each point or vector x , let $(x)_i (i = 1, 2, 3)$ denote the i -th (X, Y, Z) coordinate value of x , and $\|x\|$ denote the length of x . Let

$$\text{RMSD}(P, Q, R, \mathbf{a}) = \sqrt{\frac{1}{n} \sum_{k=1}^n \|Rp_k + \mathbf{a} - q_k\|^2}, \quad (1)$$

where R is a rotation matrix and $\mathbf{a}$ is a translation vector. Then, the *rmsd* value $d(P, Q)$ between P and Q is defined by $d(P, Q) = \min_{R, \mathbf{a}} d(P, Q, R, \mathbf{a})$. Although complicated as it might appear, the optimal rotation matrix and translation vector can be found simultaneously in $O(n)$ time. Schwartz [20] showed that $d(P, Q, R, \mathbf{a})$ is minimized when

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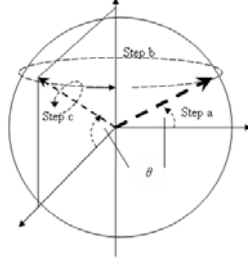


Figure 1: Three parameters (α, β, γ) , or angles $(\sin^{-1} \alpha, \beta\pi, \gamma\pi)$, suffice to determine a rigid 3D rotation of points about origin.

$\mathbf{a} = \mathbf{0}$ and

$$R = (A^t A)^{\frac{1}{2}} A^{-1}, \quad (2)$$

where the matrix $A = (A_{ij})$ $i, j = 1, 2, 3$ is given by

$$A_{ij} = \sum_{k=1}^n (p_k)_i (q_k)_j, \quad (3)$$

where $A^{\frac{1}{2}} = B$ means $BB = A$, and $\mathbf{o}$ denotes the zero vector. Thus, $d(P, Q)$, R and $\mathbf{a}$ can be computed in $O(n)$ time [17].

We adopt Martin's ProFit package (standing for protein fitting system) [16] to calculate the *rmsd* between C- α atoms of paired protein backbones. ProFit has many features including flexible specification of fitting zones and atoms, calculation of RMS over different zones or atoms, RMS-by-residue calculation. Fitting was performed using the McLachlan algorithm [17].

There have been several methods proposed to compare protein structures and measure the degree of structural similarity based on alignment of secondary structure elements as well as alignment of intra and inter-molecular atomic distances. The basic ideas are rapid identification of pair alignments of secondary structure elements, clustering them into groups, and scoring the best substructure alignment. For examples, the VAST system [6] is based on continuous distribution of domains in the fold space. The FSSP/DALI system [12] provides two levels of description – a coarse-grained one and one with a fine-grained resolution. The method, CATH, provides the complete PDB fold classification by domains and links to other sources of information. The two methods, CE and LGscore2 [21], are based on a different idea. They focus on the local geometry rather than global features such as orientation of secondary structures and overall topology (as in the case of VAST or DALI). VAST has been used to compare all known PDB domains to each other. The results of this computation are included in NCBI's Molecular Modelling Database at <http://www.ncbi.nlm.nih.gov/Structure/VAST/vast.html>.

Note that an atom-pairing scheme must be given before one can do the *rmsd* computation. The first atom of one structure is compared to the other, fifth to fifth, and so on. Our objective in this paper is to calculate the significance of *rmsd* score between spatial arrangements of C- α atom of protein backbone that are not necessarily adjacent in sequence. By matching the backbone C- α atoms between two sets of atoms, the algorithm can obtain lower *rmsd* scores comparing to these existed protein structure alignment systems like VAST or DALI.

2 Method

Our structure alignment algorithm can be decomposed into three major subroutines, ROT-M($\alpha, \beta, \gamma, \mathbf{p}$), MBM(P, Q) and STRUCALIGN($P, Q, \alpha_I, \beta_I, \gamma_I, \mathbf{p}$). The first part of our algorithm, ROT-M($\alpha, \beta, \gamma, \mathbf{p}$), produces a 3×3 rotation matrix. The matrix can be used to rotate a set of atoms to new coordinates. The MBM(P, Q) algorithm finds the minimum bipartite matching of two point sets of proteins, P and Q . STRUCALIGN($P, Q, \alpha_I, \beta_I, \gamma_I, \mathbf{p}$) is our proposed algorithm for finding a good rotation matrix and the matching function, such that two structures P, Q have a good fitting to each other. The rest of this section discusses these algorithms mentioned above in details.

2.1 ROT-M algorithm

The rotation matrix is characterized by a unit vector (probe) $\mathbf{p}$ and the 3-parameter (α, β, γ) . Here we illustrate the idea that any rotation about the origin, O , in the three-dimensional space can be expressed with a three-tuple of angles (α, β, γ) as shown in Figure 1.

Consider the point of equator-pole $\mathbf{n} = (0, 1, 0)^T$ on the unit sphere. After the rotation, R , say $\mathbf{n}$ is rotated to another point $\mathbf{q} = (x, y, z)^T$; i.e., $\mathbf{q} = R\mathbf{n}$. Let $\mathbf{q}'$ be the projection point of $\mathbf{q}$ on the yz -plane; that is, $\mathbf{q}' = (0, y, z)^T$. Let $\alpha = z$

ROT-M($\alpha, \beta, \gamma, \mathbf{p}$)

Input: (α, β, γ) are the 3-parameter controlling the rotations and $\mathbf{p}$ is a vector $(x, y, z)^T$, explained in section 2.1.

Output: A 3×3 rotation matrix M

```

1   $R_\alpha = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{1-\alpha^2} & -\alpha \\ 0 & \alpha & \sqrt{1-\alpha^2} \end{bmatrix}$   $\triangleright$  rotate around  $x$ -axis by angle  $\sin^{-1} \alpha$ .
2   $R_\beta = \begin{bmatrix} \cos(\beta\pi) & -\sin(\beta\pi) & 0 \\ \sin(\beta\pi) & \cos(\beta\pi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$   $\triangleright$  rotate around  $z$ -axis by angle  $\beta\pi$ .
3   $(x, y, z) \leftarrow R_\beta \cdot R_\alpha \cdot \mathbf{p}$ ;  $c \leftarrow \cos(\gamma\pi)$ ;  $s \leftarrow \sin(\gamma\pi)$ .
4   $R_\gamma = \begin{bmatrix} c + x^2 \cdot (1-c) & xy \cdot (1-c) - zs & xz \cdot (1-c) + ys \\ xy \cdot (1-c) + zs & c + y^2 \cdot (1-c) & yz \cdot (1-c) - xs \\ xz \cdot (1-c) - ys & yz \cdot (1-c) + xs & c - z^2 \cdot (1-c) \end{bmatrix}$   $\triangleright$  rotate around  $(x, y, z)^T$  by angle  $\gamma\pi$ .
5  return  $R_\gamma \cdot R_\beta \cdot R_\alpha$ 

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Figure 2: ROT-M algorithm for 3 parameters.

and thus $\theta = \sin^{-1} \alpha$ is the angle $\angle \mathbf{nOq}'$. Note that $\alpha = \sin \theta$ is exactly the z -coordinate of $\mathbf{q}'$. To determine x -coordinate and y -coordinate of $\mathbf{q}$, the point is rotated around the z -axis by the angle β . The particular rotation R can be decided by rotating all other points around the vector $\mathbf{q}$ by the angle γ . It is not hard to verify that, in such a way, *any* rigid rotation transformation can be parameterized by the triple (α, β, γ) or (z, β, γ) .

Thus, we call a vector $\mathbf{p} = (x, y, z)^T$ on the surface of the unit sphere a *probe*. Note that the vector $\mathbf{p}$ is rotated to the probe, $\mathbf{p}''$, on the unit sphere. The position of $\mathbf{p}''$ is decided by the parameters α and β , and the final transformation is to rotate points around $\mathbf{p}''$ by angle γ .

To uniformly place several probes on surface of the sphere, we consider the distribution function of the location of a probe $\mathbf{p}$ on the sphere in terms of the two parameters α and β , as illustrated in Figure 1. Note that β is clearly uniformly distributed in the range $(-\pi, \pi)$. On the other hand, the upper angle, $\theta = \sin^{-1} \alpha$, is not. Let $f(x)$ be the probability density function of a random variable $X: \mathbb{R} \rightarrow \mathbb{R}$, and let F be the probability cumulative function $F(x) = \int_{-\infty}^x f(t)dt$. It can be verified that $x = F^{-1}(p)$, where p is taken uniformly from $(0, 1)$, is a reasonable way of generating a random sample x that matches the desired probability distribution defined by f . Furthermore, it is not hard to verify that $f(\theta) \propto \cos \theta$; thus $F(\theta) = \sin \theta$. It follows that $\theta = \sin^{-1}(\text{RAND}(-1, 1))$, is the way of generating a good uniformly distributed random sample point. It is interesting to note that, since $\alpha = \sin \theta$, we can just set $\alpha = \text{RAND}(-1, 1)$ to be the desired distribution.

That is, our 3-parameter method (α, β, γ) can be summarized as the following:

- **Rotation around x -axis:**

Given a unit vector $\mathbf{p} = (x, y, z)^T$, $\mathbf{p}$ is transformed into $\mathbf{p}'$ by a rotation around the x -axis by angle $\sin^{-1} \alpha = \theta$. That is, let

$$\mathbf{p}' = \begin{pmatrix} x_\alpha \\ y_\alpha \\ z_\alpha \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{1-\alpha^2} & -\alpha \\ 0 & \alpha & \sqrt{1-\alpha^2} \end{bmatrix} \cdot \mathbf{p}$$

Since $\sin \theta = \alpha$ and thus, $\cos \theta = \sqrt{1-\alpha^2}$.

- **Rotation around z -axis:**

The vector, $\mathbf{p}' = (x_\alpha, y_\alpha, z_\alpha)^T$, is transformed into the probe $\mathbf{p}''$ by a rotation around the z -axis by angle $\beta\pi$. That is, let

$$\mathbf{p}'' = \begin{pmatrix} x_\beta \\ y_\beta \\ z_\beta \end{pmatrix} = \begin{bmatrix} \cos \beta\pi & -\sin \beta\pi & 0 \\ \sin \beta\pi & \cos \beta\pi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \mathbf{p}'$$

then we will get new coordinate of $(x_\beta, y_\beta, z_\beta)^T$.

- **Rotation around the probe $\mathbf{p}''$:**

The last rotation matrix, R_γ , do the body rotation around the probe $\mathbf{p}''$ by angle $\gamma\pi$; see [9] for related discussions about the transformation. That is, let

$$(x, y, z) = (x_\beta, y_\beta, z_\beta)^T.$$

$$c = \cos \gamma\pi, s = \sin \gamma\pi, h = 1 - c.$$

$$R_\gamma = \begin{bmatrix} c + x^2h & xyh - zs & xzh + ys \\ xyh + zs & c + y^2h & yzh - xs \\ xzh - ys & yzh + xs & c - z^2h \end{bmatrix}$$

The final concise form of the algorithm is show in Figure 2.

Euler's Rotation Theorem

According to Euler's rotation theorem [14], any rotation can about the origin point be described by using three angles. The rotation is determined by 3 consecutive rotations with 3 *Euler angles* $(\theta, \beta', \gamma')$. The first rotation is done by the angle θ ($= \sin^{-1} \alpha$) around the z -axis, the second is done by the angle β' ($= \beta\pi$) around the x -axis, and the third rotation is done by the angle γ' around the z -axis.

Here we explain the reason that we choose our method instead of Euler's 3-parameter method. Note that the $\alpha = z$ is just $\sin \theta$, where θ is Euler's first angle. The obvious advantage of using the parameter α is letting the system directly obtain values $\sin \theta$ and $\cos \theta$. Most importantly, the probability distribution of the searching space of probes is evenly distributed on the surface of a unit sphere by the $\alpha = \sin \theta = \text{RAND}(-1,1)$ method, as mentioned in Section 2.1. Also note that Euler's third rotation is to rotate around z -axis by angle γ' ; on the other hand, our method is to rotate around the probe vector by the angle $\gamma\pi$.

As a result, we reduce the problem of finding a good rotation matrix to the new problem of finding a good 3-parameter. The rotation matrix is thus characterized by just adjusting the 3 uniformly distributed parameters. We show this algorithm in Figure 2.

2.2 MBM algorithm

The main idea of our algorithm is to search for the suitable 3-parameter, which leads to a suitable matching between two sets of points. After the matching list is found, we then use the RMSD procedure to find the suitable rotation transformation. Given two sets of points P and Q with the rotation transformation R , let $P' = R \cdot P$, and let Q be translated to Q' such that the mass center of Q' is located at the origin. We construct a weighed graph $G = (V, E)$ with V being labelled with points of P' and Q' , and each (p, q) in E being weighted with the squared Euclidean (3D) distance; i.e., $w(p, q) = \|p, q\|^2$. We then solve the weighted minimum bipartite matching problem [8] to obtain the best matching of P' and Q' . By the matched pairing, we perturb and refine the final alignment to obtain lower *rmsd*; for brevity of presentation, the detailed description is sketched in Figure 3.

There are several proposed algorithms for the minimum bipartite matching problem; sometimes it is also referred as the *assignment problem*. Here we adopted the Munkres [18, 2, 1, 19] algorithm. The public available implementation is written with Perl language. To improve the efficiency of computation, we implement the Munkres algorithm and write hundreds lines of C Codes, and

and the produced codes are strictly verified by comparing the results with the public Perl package. The resulting C-coded program is about 3 times faster than the Perl version system.

2.3 STRUC-ALIGN algorithm

The main idea behind our STRUC-ALIGN algorithm is the parametric searching for a good 3-parameter. The suitable 3-parameter determines a good rotation transformation R , and the rotation R , after the minimum bipartite matching procedure, then leads to a good alignment of two protein structures. Assuming the initial probe position $\mathbf{p}$ is fixed, the 3-parameter would eventually leads to a particular *rmsd* value s , which in turn serves as an measurement function for the 3-parameter, as shown in line 8-9 of STRUC-ALIGN algorithm of Figure 3. The goal is to find the 3-parameter (and its corresponding rotation transformation) which leads to the smallest possible *rmsd* value.

The idea of STRUC-ALIGN algorithm is to iteratively explore the proximity of a probe over the searching space in a randomized greedy manner. Thus, the PERTURB subroutine of STRUC-ALIGN algorithm is the uniformly distributed random variance of 3-parameter as shown in Figure 3. Let (α, β, γ) be the current probe. Note that $(\alpha_{max}, \beta_{max}, \gamma_{max})$ is the maximal possible range of the 3-parameter. The PERTURB subroutine *shacks* the probe (α, β, γ) to a neighboring position around its proximity. Since the best parametric probe will be very difficult to locate, so as to avoid unnecessary redundancy of computation, the total searching times must be limited by as many as T trials number as shown in the line 5 of STRUC-ALIGN algorithm of Figure 3.

We explain the line 9-11 of STRUC-ALIGN algorithm as follow. The variable s represents the best *rmsd* value found by the algorithm at present. If the perturbed measurement s' is less than s , it means the algorithm finds a better 3-parameter position. So we let the perturbed probe $(\alpha', \beta', \gamma')$ substitute the current probe position, directing the PERTURB subroutine started from the new position. On the other hand, if there are consecutive F attempts unable to locate better neighboring probes, the line 11 of STRUC-ALIGN algorithm shows that it will restart the whole searching from the initial position $(\alpha_I, \beta_I, \gamma_I)$.

2.4 The main algorithm

In order to obtain the best possible alignment of two given protein structures, the STRUC-ALIGN algorithm relies on a *good* initial probe position $(\alpha_I, \beta_I, \gamma_I)$. Since the good initial probe will be

STRUC-ALIGN($P, Q, \alpha_I, \beta_I, \gamma_I, \mathbf{p}$)

Input: Two set of 3D coordinates of points $P = \{p_1, p_2, \dots, p_n\}$ and $Q = \{q_1, q_2, \dots, q_m\}$; $n < m$.

The α_I , β_I and γ_I are real numbers that are between -1 to 1.

▷ These inputs control the initial position of 3 parameters box and affect the explored area.

▷ $\mathbf{p}$ is the vector $(x, y, z)^T$, explained in section 2.1.

Output: $(s, \alpha, \beta, \gamma)$ is a sufficiently low RMSD s and (α, β, γ) .

▷ (α, β, γ) is the best position of 3 parameters box.

Global : $T, F, \alpha_{max}, \beta_{max}, \gamma_{max}$.

The threshold T and F are both integer numbers.

▷ T is total trial number of perturbation.

▷ F is the maximum number of consecutive failed perturbations for a probe to restart.

$\alpha_{max}, \beta_{max}, \gamma_{max}$ are real numbers between 0 to 1.

▷ These inputs control the range of parameters perturbation variances.

1 $(\alpha, \beta, \gamma) \leftarrow (\alpha_s, \beta_s, \gamma_s) \leftarrow (\alpha_I, \beta_I, \gamma_I)$.

2 $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha, \beta, \gamma, \mathbf{p}))$. ▷ Q' is a temp array of atoms set of protein.

3 $L \leftarrow \text{MBM}(P, Q')$; $(R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q')$; $s \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$.

4 $f \leftarrow 0; t \leftarrow 0$. ▷ Initializing the counter.

5 **while** $t \leq T$ **do**

6 $(\alpha', \beta', \gamma') \leftarrow \text{PERTURB}(\alpha, \beta, \gamma)$.

7 $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha', \beta', \gamma', \mathbf{p}))$. ▷ Q' is a temp array of atoms set of protein.

8 $L \leftarrow \text{MBM}(P, Q')$; $(R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q')$; $s' \leftarrow \text{RMSD}(P, Q', R, \mathbf{a})$.

9 **if** $s' \leq s$ **then** $s \leftarrow s'$; $(\alpha, \beta, \gamma) \leftarrow (\alpha_s, \beta_s, \gamma_s) \leftarrow (\alpha', \beta', \gamma')$; $f \leftarrow 0$;

10 **else** $f \leftarrow f+1$.

11 **if** $f \geq F$ **then** $(\alpha, \beta, \gamma) \leftarrow (\alpha_I, \beta_I, \gamma_I)$; $f \leftarrow 0$.

12 $t \leftarrow t+1$.

13 **return** $(s, \alpha_s, \beta_s, \gamma_s)$.

MBM(P, Q) returns the minimum bipartite matching of two point sets P and Q .

PERTURB(α, β, γ).

Input: The α, β and γ are real numbers between -1 to 1.

▷ These inputs control the present position of 3 parameters box and affect the explore area.

Output: 3 real numbers $(\alpha', \beta', \gamma')$.

▷ These outputs are the new position of 3 parameters box.

1 $\Delta\alpha \leftarrow \text{RAND}(-\alpha_{max}, \alpha_{max})$; $\Delta\beta \leftarrow \text{RAND}(-\beta_{max}, \beta_{max})$; $\Delta\gamma \leftarrow \text{RAND}(-\gamma_{max}, \gamma_{max})$.

2 $\alpha' \leftarrow \text{BACK}(\alpha + \Delta\alpha)$; $\beta' \leftarrow \text{ROUND}(\beta + \Delta\beta)$; $\gamma' \leftarrow \text{ROUND}(\gamma + \Delta\gamma)$.

3 **return** $(\alpha', \beta', \gamma')$.

RAND(a, b) is a random function returning a real number uniformly distributed between a and b .

$$\text{BACK}(\alpha) = \begin{cases} -2 - \alpha & \text{if } \alpha \leq -1 \\ 2 - \alpha & \text{if } \alpha \geq 1 \\ \alpha & \text{otherwise} \end{cases} ; \text{ROUND}(\theta) = \begin{cases} 2 + \theta & \text{if } \theta \leq -1 \\ -2 + \theta & \text{if } \theta \geq 1 \\ \theta & \text{otherwise} \end{cases}$$

TRANS(A, R).

Input: A is an array of 3D points with size n .

R is the rotation matrix.

Output: An array of 3D points, B .

1 **for** $i \leftarrow 1$ to n **do**

2 $B[i] \leftarrow R \cdot A[i]$ ▷ B is the array containing the transformed n points.

3 **return** B .

Figure 3: Aligning two sets of atoms with low *rmsd* by pairing points according to the minimum bipartite matching measurement

ALIGN-RANDOM($P, Q, T, F, \alpha_{max}, \beta_{max}, \gamma_{max}, N$)

Output: (s, L)

```

1   $s \leftarrow \infty$  ;  $\mathbf{p} \leftarrow (0, 1, 0)^T$ .
2  for  $k \leftarrow 1$  to  $N$ 
3       $\alpha \leftarrow \text{RAND}(-1, 1)$  ;  $\beta \leftarrow \text{RAND}(-1, 1)$  ;  $\gamma \leftarrow \text{RAND}(-1, 1)$ .
4       $\triangleright (\alpha, \beta, \gamma)$  are uniformly distributed random values ranged in  $(-1, 1)$ .
5       $(s', \alpha'_s, \beta'_s, \gamma'_s) \leftarrow \text{STRUC-ALIGN}(P, Q, \alpha, \beta, \gamma, \mathbf{p})$ .
6      if  $s' \leq s$  then  $s \leftarrow s'$  ;  $(\alpha_s, \beta_s, \gamma_s) \leftarrow (\alpha'_s, \beta'_s, \gamma'_s)$ .
7   $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha_s, \beta_s, \gamma_s, \mathbf{p}))$ .  $\triangleright Q'$  is a temp array of atoms set of protein.
8   $L \leftarrow \text{MBM}(P, Q')$ .  $\triangleright L$  is the matching list of point sets  $P$  and  $Q$ .
9  return  $(s, L)$ .
```

Figure 4: Initial random seed.

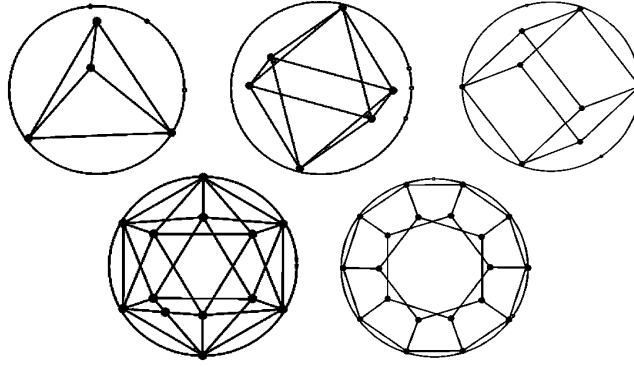


Figure 5: The fixed initial setting of the system, where the probes are distributed fixedly and equally on the sphere.

ALIGN-UNIFORM($P, Q, T, F, \alpha_{max}, \beta_{max}, \gamma_{max}, N$)

Output: (s, L)

```

1   $s \leftarrow \infty$ .
2   $U \leftarrow \text{INIT-S}(N)$ .
3       $\triangleright$  Placing seeds by  $\text{INIT-S}(N)$ , and number of seeds determined by  $N$ .
4  for each  $u \in U$  do
5       $\mathbf{p} \leftarrow u$  ;  $(s', \alpha'_s, \beta'_s, \gamma'_s) \leftarrow \text{STRUC-ALIGN}(P, Q, 0, 0, 0.5, \mathbf{p})$ .
6      if  $s' \leq s$  then  $s \leftarrow s'$  ;  $(\alpha_s, \beta_s, \gamma_s) \leftarrow (\alpha'_s, \beta'_s, \gamma'_s)$ .
7   $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha_s, \beta_s, \gamma_s, \mathbf{p}))$ .  $\triangleright Q'$  is a temp array of atoms set of protein.
8   $L \leftarrow \text{MBM}(P, Q')$ .  $\triangleright L$  is the matching list of point sets  $P$  and  $Q$ .
9  return  $(s, L)$ .
```

Figure 6: Initial fixed seed.

ALIGN-REFINE($P, Q, T, F, \alpha_{max}, \beta_{max}, \gamma_{max}, L$)

$\triangleright L$ is the given alignment list (usually obtained from other method.)

Output: (s, L)

```

1   $(R, \mathbf{a}) \leftarrow \text{MS-FIT}(L, P, Q)$ .
2   $P \leftarrow \text{TRANS}(P + \mathbf{a}, R)$ .  $\triangleright$  Rotating the points of  $P$  to the  $(0, 0, 0)$  probe position.
3   $\mathbf{p} \leftarrow (0, 1, 0)^T$  ;  $(s, \alpha_s, \beta_s, \gamma_s) \leftarrow \text{STRUC-ALIGN}(P, Q, 0, 0, 0, \mathbf{p})$ .
4   $Q' \leftarrow \text{TRANS}(Q, \text{ROT-M}(\alpha_s, \beta_s, \gamma_s, \mathbf{p}))$ .  $\triangleright Q'$  is a temp array of atoms set of protein.
5   $L \leftarrow \text{MBM}(P, Q')$ .  $\triangleright L$  is the matching list of point sets  $P$  and  $Q$ .
6  return  $(s, L)$ .
```

Figure 7: Refinement of a given structure alignment.

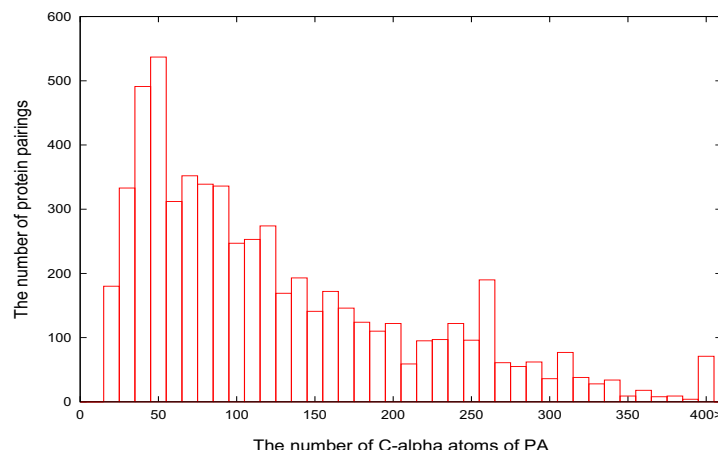


Figure 8: The distribution of the 200 randomly picked protein structures in PDB and their 30 neighbor structures. The total number of protein pairs is 6,000.

very difficult to locate, the idea is to send out a team of several probes over the sphere and let each probe searching its own proximity in a randomized greedy manner. To do so, the ALIGN-RANDOM algorithm, as shown in Figure 4, is to set N randomized probes $(\alpha_I, \beta_I, \gamma_I)$, with each probe being selected randomly from the uniformly distributed 3-parameter box.

On the other hand, the ALIGN-UNIFORM algorithm is to set $(\alpha_I, \beta_I, \gamma_I)$ evenly distributed over the probe searching space in a pre-calculated manner. Note in ROT-M algorithm, the rotation matrix is effected by both the 3-parameter and a given initial probe position. Thus, we use the idea of *regular polyhedron* on the 3D sphere (or also known as the *Platonic solids*) [5], whose vertices are placed evenly over the surface of a unit sphere, as illustrated by Figure 5. Each point of the regular polyhedra is our initialized vector $\mathbf{p}$, shown in the line 2-4 of ALIGN-UNIFORM algorithm of Figure 6. Note the initiatory γ is 0.5, as shown in the line 5 of ALIGN-UNIFORM algorithm of Figure 6. Note that the γ value can not be set to 0; otherwise, the corresponding rotation matrix is always the unit matrix, leading to an identity transformation, regardless of any setting of probe position.

Refinement of a given alignment

It is interesting to note that it is very easy to adopt the the STRUC-ALIGN algorithm to improve *any* given structure alignment, as shown by ALIGN-REFINE illustrated in Figure 7. The idea behind this algorithm is for testing and verifying the practicability of our method. If and only if, after we refine the VAST alignment and find a better alignment, then we enable to understand how much better that our method is. As shown in the line

1-2 of ALIGN-REFINE algorithm of Figure 7,

Suppose we are given an existed structure alignment (usually obtained by other protein structure alignment methods, e.g., say the VAST package.) The ALIGN-REFINE algorithm use this initial alignment to get the suitable rotation matrix. The matrix is then used to rotate the coordinates of atoms of one protein, as shown in the line 1-2 of Figure 7. Line 3 of ALIGN-REFINE then use the STRUC-ALIGN algorithm to improve the given structure alignment.

3 Experimental Result

In this section, we show the experimental method and result. First, we introduce the available protein structure database and how we choose the desired samples for our experiment. Secondly, we show how the data is collected and our protein pairings method. In the third part, we analyze the improvement ratios of our method tested on the given data. Finally, we analyze the experimental execution time of our programs.

3.1 Sample Source

Among various public domain providers of protein structures, two protein structure databases are well known and considered most reliable. One is PDB (Protein Data Bank) [10], and the other is MMDB (Molecular Modeling DataBase) [11]. The RCSB PDB provides a variety of tools and resources for studying the structures of biological macromolecules and their relationships to sequence, function, and disease. The MMDB is NCBI's structure database. It is a subset of three-dimensional structures obtained from the Protein Data Bank (PDB), excluding theoretical models.

MMDB is a database of ASN.1-formatted records. It was designed for flexibility, and as such, is capable of archiving conventional structural data as well as future descriptions of biomolecules, such as those generated by electron microscopy (surface models).

We choose the PDB for our experimental sample source. The RCSB PDB is the main provider of PDB files. It is supported by funds from the National Science Foundation (NSF), the National Institute of General Medical Sciences (NIGMS), the Office of Science, Department of Energy (DOE), and so on. The RCSB PDB is most commonly referred by *J.Mol.Biol.*, *Biochemistry*, *J.Biol.Chem.*, and other well known bioinformatics literatures. To date, there are more than 40,000 protein structures listed in the PDB database. For its abundance of data collection, we use this data set for our experimental population.

In our experiment, we randomly pick 200 protein structures in the PDB database as our experimental subjects by the uniform distribution sampling. Each chosen structure must satisfy two conditions. First, there must be more than 100 structure neighbors in the VAST web site. Secondly, the alignment in VAST must be the same as the sequence in the PDB file. If the chosen structure satisfies the two conditions, then we randomly choose 30 structure alignments listed on the database as the tested targets. Totally, there are 6,000 protein pairs tested by our experiment. We use the term, PA, to stand for one of the 200 randomly picked protein structures, and we use PB to stand for one of the 30 neighbors of each PA.

We illustrate the relation between the number of C- α atoms of PA, and the number of protein pairs at Figure 8. It is very clear that most of the samples are centralized from 20 to 200. All together, the total number of protein pairs is 6,000.

3.2 Data Collection Method

For each randomly picked PDB structure, we obtain its structure neighbors in the NCBI's VAST web site. Then, a set of perl programs is used to analyze the web pages and obtain all structure alignments between the chosen PDB structure and its structure neighbors. We found that the alignment in VAST is not always the same as the sequence of PDB file. Therefore, we need to use the structure alignment to test against the sequence of the PDB file to make sure that the PDB is suitable for using in our experiment. According to the structure alignment, given by the VAST, we analyze the two PDB files and get the sequences of C- α atoms. The original VAST alignment of (PA, PB) protein pairs is improved by our algorithm described in Figure 7 ALIGN-REFINE. An example

is illustrated at Figure 10. Note that the ordering of atoms could be changed or even reversed after the refined structure alignment process.

3.3 Experimental Improvement Ratios

Our proposed method first measures each sampled pair by calculating its original alignment's *rmsd* value. The measurement is confirmed and double checked with the *rmsd* by the *PROFIT* package kits [16].

The results show that our proposed method improves almost every sampled VAST pairs. By the ascending order on the number of C- α atoms of PA, we partition the samples into 40 points, with each point standing for 150 protein pairs. Figure 11 shows the relation between the number of C- α atoms of PA and the improvement ratios contrast to the given VAST alignment. The formula of the improvement ratio is defined by

$$\rho = (A - B)/A,$$

where A is the original *rmsd* value by the VAST alignment, while B is improved(smaller) *rmsd* value by our structure alignment method. The improvement ratios of our samples are mostly distributed from 5% to 15%. As a total, the average of the improvement ratios is 9.29%.

3.4 The Execution Time of the Experiment

The execution time of the experiment for our proposed method is summarized in Figure 12, which shows the relation between the number of C- α atoms of PA and the execution time (CPU processor time) of our structure alignment system needed for the samples to run. Note that each drawn point in Figure 12 represents a group of 150 protein pairs. The structure alignment system is implemented and tested under the Linux Red Hat 4.0 system. Each of the experimental machine is equipped with AMD Athlon(tm) XP 2400+ processor and 1GB RAM main memory. The experiment is performed in parallel on 11 machines, and it takes approximately 70 hours or 3 days to finish the experiment on the total 6,000 protein structure pairings.

4 Conclusion and Future Work

Bioinformatics has become an essential tool not only for basic research but also for applied research in biotechnology and biomedical sciences. Currently, the field is under going an enormous expansion, as witnessed by the dramatic increase in the number of related bioinformatic literatures.

In this paper, we propose algorithms which improve the *rmsd* value between a protein structure pair by finding better alignment list. As it is shown by the experiment results, our method improves the alignment computed by the VAST package by an averaged improvement ratios about 9.29%. So far, the system demonstrates that the method of using 3D Euclidean distance minimum bipartite matching with perturbed parametric searching scheme indeed improves existed known system like the VAST. It will be interesting to know how well our algorithms performs comparing to other existed known system, e.g. CE, DALI.

Parts of our preliminary experiment results have been incorporated into our web-based system, and the system is accessible at <http://bioinfo.cs.pu.edu.tw/~pco/pcomp.html>. Efforts are been made to build up a more informative and friendly web-based system to provide interested molecular biologists to test their interested protein structure pairing. We hope the system will be ready for practical use soon. Furthermore, since the structure comparison problem, like many scientific computation/simulation problem, is very time-consuming under cases of large structures and large number of paired structures, it is desirable to implement the system under massive parallel machines cluster, e.g., the grid-environment, to increase the throughput of the system.

There is an important direction of research along the topic of protein structure alignment. The problem of *local structure alignment* is to find the functional (or active) part of a given query protein; these active parts imply a substructure similarity between two proteins. The problem of identification of the similar substructure of a protein pair will indeed be a challenge.

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The number of C- α atoms of PA	The number of protein pairings	The average of improved ratio (%)	The average of execution time (Sec)	The standard deviation of improved ratio (%)	The standard deviation of execution time (Sec)
50<	1541	7.61	0.57	11.99	1.27
50-100	1586	7.17	5.98	11.98	5.54
100-150	1030	8.42	15.16	12.48	14.83
150-200	674	14.71	41.73	15.61	41.49
200-250	469	11.30	161.01	14.74	160.92
250-300	404	9.57	401.65	15.07	401.90
300-350	186	17.57	873.34	18.21	875.59
350-400	40	12.66	1592.63	25.30	1612.83
400>	70	15.67	10315.05	21.41	10389.45
Total	6000	9.29	206.68	13.22	206.26

Figure 9: The distribution of our experimental data containing the number of C- α atoms of PA, the number of protein pairings, the average of improved ratio, the average of execution time, the standard deviation of improved ratio and the standard deviation of execution time.

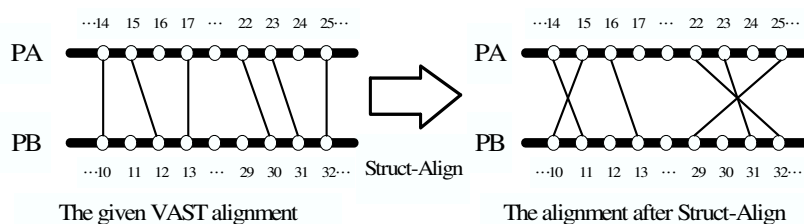


Figure 10: An example of protein structures alignment before and after the minimum bipartite matching.

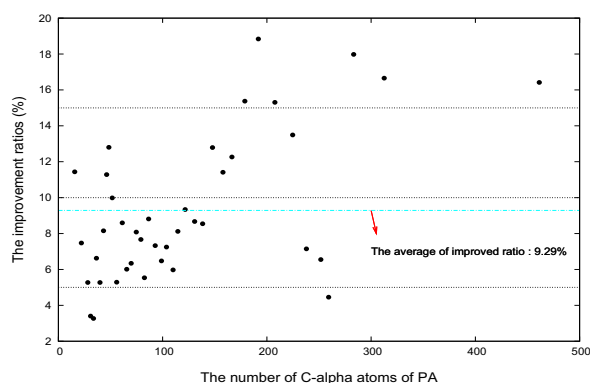


Figure 11: The improvement ratios opposite to the *rmsd* of the VAST alignment. The improvement ratios are mostly distributed from 5% to 15%. The average is 9.29%.

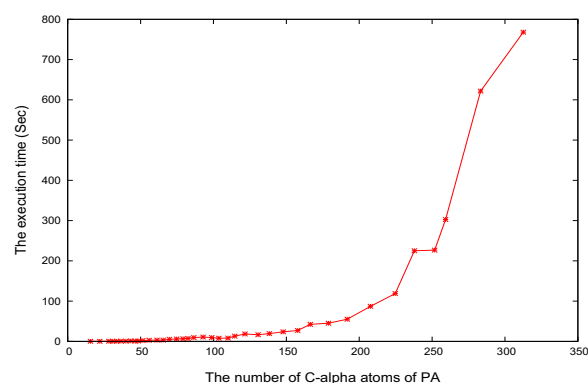


Figure 12: The execution time of our structure alignment system needed for the samples to run.

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